

Sangola College, Sangola
Dept. of Computer Science & Applications
B.Sc(ECS)-III
(Data Mining using weka)
ASSIGNMENT No-01

Date:

Create a employee.csv dataset. Solve (Write a procedure) following questions with the help of Data Mining Tool WEKA.

EmpID	Ename	Department	Gender	Age	Salary	Experience	Performance
201	Amit	IT	Male	24	35000	2	High
202	Priya	HR	Female	27	40000	3	Medium
203	Ramesh	Sales	Male	30	?	5	Low
204	Sneha	IT	Female	26	45000	?	High
205	Kunal	Finance	Male	29	48000	4	Medium
206	Meena	HR	Female	28	42000	3	High
207	Rajesh	IT	Male	31	50000	6	Low
208	Shital	Sales	Female	25	36000	2	?
209	Vinay	Finance	Male	?	55000	7	Medium
210	Kavita	HR	Female	23	38000	1	High

- 1) Load the dataset into WEKA.
- 2) Add a new attribute "Grade" (Nominal: {A, B, C}) at index 6.
- 3) List all attributes in the dataset.
- 4) Identify which attributes have missing values and calculate their percentage.
- 5) Remove any attribute if missing values are more than 40%.
- 6) Convert the EmpID attribute from numeric to nominal.
- 7) Replace missing values (use ReplaceMissingValues filter).
- 8) Discretize the Salary attribute into 3 bins (Low, Medium, High).
- 9) Copy the Experience attribute to the 4th position.
- 10) Detect and remove any outliers in the Age attribute.
- 11) Normalize the Salary attribute using Normalize filter.
- 12) Standardize the Age attribute.
- 13) Remove duplicate records.
- 14) Apply Attribute Selection and list the best subset of features.
- 15) Replace all missing numeric values with 0 and nominal values with "Unknown".
- 16) Retrieve all instances where Salary > 45000.
- 17) Remove 25% of instances using the RemovePercentage filter.
- 18) Produce a 20% random resample of the dataset.
- 19) Change the attribute name Ename → Employee_Name.
- 20) Save the processed dataset as Employee_processed.arff.